Home / Grade 6 / Geometry / Quarter 2

Multiplying and dividing by powers of ten

A Cambridge insert: the digits move, the point stays — and why that is the better picture.

LESSON 56–57 | 2 × 40 MIN | MATEMATIKA 6, QO‘SHIMCHA MAVZU | S7 3.1

LESSON CLOCK

Place value	20 min
Multiplying and dividing by *10, 100, 1000*	25 min
By *0.1* and *0.01*	25 min
Converting units	8 min
Homework	2 min

LEARNING OBJECTIVES

- Multiply and divide by *10*, *100* and *1000*.
- Multiply and divide by *0.1* and *0.01*.
- Explain the effect in terms of place value.
- Use the rules to convert between units.

TEXTBOOK

National	pp. 158–163
Cambridge	Stage 7, pp. 32–36
Workbook	Exercise 3.1

01 Explanation

Place value

COLUMN	HUNDREDS	TENS	UNITS	TENTHS	HUNDREDTHS
value	*100*	*10*	*1*	$\frac{1}{10}$	$\frac{1}{100}$
3.47	—	—	*3*	*4*	*7*
347	*3*	*4*	*7*	—	—

EACH COLUMN IS TEN TIMES THE ONE ON ITS RIGHT

Multiplying by *10* moves every digit one column to the left, which is the same thing as moving the point one place to the right. The first description is the true one; the second is a shortcut.

Multiplying and dividing by *10, 100, 1000*

START	$\times 10$	$\times 100$	$\div 10$	$\div 100$
3.47	34.7	347	0.347	0.0347
58	580	5800	5.8	0.58
0.6	6	60	0.06	0.006

ZEROS HOLD THE EMPTY COLUMNS

$0.347 \div 10$ is 0.0347: a zero has to appear in the tenths column, because that column is now empty. Leaving it out changes the number tenfold.

By 0.1 and 0.01

Multiplying by 0.1 is the same as dividing by 10 — and dividing by 0.1 is the same as multiplying by 10.

OPERATION	SAME AS	EXAMPLE
$\times 0.1$	$\div 10$	$47 \cdot 0.1 = 4.7$
$\times 0.01$	$\div 100$	$47 \cdot 0.01 = 0.47$
$\div 0.1$	$\times 10$	$47 \div 0.1 = 470$
$\div 0.01$	$\times 100$	$47 \div 0.01 = 4700$

MULTIPLYING CAN MAKE A NUMBER SMALLER

$47 \cdot 0.1 = 4.7$. “Multiplication makes bigger” is only true for multipliers above 1 — one of the most useful facts of the year to unlearn.

Converting units

CONVERSION	OPERATION	EXAMPLE
m to cm	$\times 100$	$3.5\ m = 350\ cm$
cm to m	$\div 100$	$47\ cm = 0.47\ m$
km to m	$\times 1000$	$2.4\ km = 2400\ m$
g to kg	$\div 1000$	$750\ g = 0.75\ kg$
l to ml	$\times 1000$	$1.5\ l = 1500\ ml$

SMALLER UNIT, BIGGER NUMBER

A metre is many centimetres, so the number of centimetres is larger. Deciding whether to multiply or divide is a question about the units, not about the digits.

02 Worked examples

EXAMPLE 1 **Compute $3.47 \cdot 100$ and $3.47 \div 100$.**

1. Each digit moves two columns left: 347.
2. Each digit moves two columns right: 0.0347.

   Zeros fill the empty columns.

ANSWER

347 and 0.0347

EXAMPLE 2 **Compute $47 \div 0.1$.**

1. Dividing by 0.1 is multiplying by 10.
2. How many tenths are in 47? Ten in each unit.
3. 470

ANSWER

470

EXAMPLE 3 **Convert 750 g to kilograms and 2.4 km to metres.**

1. A kilogram is 1000 g, so divide: 0.75 kg.
2. A kilometre is 1000 m, so multiply: 2400 m.

ANSWER

0.75 kg and 2400 m

03 Interactive model

Write the place-value columns once on the board and move the digits along it rather than the point; the zeros then appear where they are needed without a rule.

Which way do the digits move?

$3.47 \cdot 10$ equals:

- 34.7
- 0.347
- 347
- 3.470

$3.47 \div 100$ equals:

- 0.347
- 0.0347
- 347
- 34.7

$58 \div 10$ equals:

- 5.8
- 0.58
- 580
- 5800

$47 \cdot 0.1$ equals:

- 470
- 4.7
- 0.47
- 4700

$47 \div 0.1$ equals:

- 4.7
- 470
- 0.47
- 4700

$47 \div 0.01$ equals:

- 470
- 4700
- 0.47
- 4.7

3.5 m in centimetres:

- 35
- 350
- 0.035
- 3500

750 g in kilograms:

- *7.5*
- *0.75*
- *75*
- *0.075*

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Place value	Xona qiymati	Разрядное значение
Power of ten	O‘nning darajasi	Степень десяти
Decimal point	O‘nli nuqta (vergul)	Десятичная запятая
To shift	Siljimoq	Сдвигаться
Tenth	O‘ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Zero as a placeholder	O‘rin to‘ldiruvchi nol	Ноль как разделитель
Unit conversion	Birliklarni almashtirish	Перевод единиц

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Multiplying by 10 moves each digit:

- one column left
- one column right
- two left
- nowhere

Dividing by 100 moves each digit:

- two left
- two right
- one right
- nowhere

Multiplying by 0.1 is the same as:

- dividing by 10
- multiplying by 10
- adding 0.1
- nothing

Dividing by 0.01 is the same as:

- dividing by 100
- multiplying by 100
- multiplying by 0.01
- nothing

$0.347 \div 10$ equals:

- 0.0347
- 0.347
- 3.47
- 0.00347

To change metres to centimetres:

- divide by 100
- multiply by 100
- multiply by 10
- divide by 10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3.47 \cdot 10$

   ANSWER 34.7

2. $3.47 \cdot 100$

   ANSWER 347

3. $3.47 \div 10$

   ANSWER 0.347

4. $3.47 \div 100$

   ANSWER 0.0347

5. $58 \cdot 100$

   ANSWER 5800

6. $58 \div 10$

   ANSWER 5.8

7. $0.6 \cdot 100$

   ANSWER 60

Medium · the standard question

7 PROBLEMS

1. $47 \cdot 0.1$

   ANSWER 4.7

2. $47 \cdot 0.01$

   ANSWER 0.47

3. $47 \div 0.1$

   ANSWER 470

4. $47 \div 0.01$

   ANSWER 4700

5. 3.5 m in cm

   ANSWER 350

6. 47 cm in m

   ANSWER 0.47

7. 750 g in kg

   ANSWER 0.75

Hard · stretch and reason

7 PROBLEMS

1. $0.06 \div 0.1$

   ANSWER 0.6

2. $0.06 \cdot 1000$

   ANSWER 60

3. 2.4 km in cm

   ANSWER $240\,000$

4. 1.5 l in ml

   ANSWER 1500

5. $0.008 \cdot 100$

   ANSWER 0.8

6. Why does $47 \cdot 0.1$ make the number smaller?

   ANSWER The multiplier is less than one

7. $3\,200$ ml in litres

   ANSWER 3.2

07 Homework

Homework — 5 tasks

Ask first whether the answer should be bigger or smaller; that settles the direction.

1. Compute $5.28 \cdot 10$, $5.28 \cdot 1000$ and $5.28 \div 100$.
2. Compute $36 \cdot 0.1$ and $36 \div 0.01$.
3. Convert 4.7 m to centimetres and 86 cm to metres.
4. Convert 3.6 kg to grams and 450 ml to litres.
5. Explain in one sentence why $\div\ 0.1$ makes a number larger.